

# Predictive resilience assessment of road networks based on dynamic multi-granularity graph neural network

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## ABSTRACT

Due to the influence of global climate anomalies, abnormal weather conditions such as heavy rainfall have become more frequent in recent years, posing a significant threat to the operation of transportation systems. An effective assessment of the resilience of the transportation system before and during heavy rain can alert the transportation department to take necessary emergency actions. However, existing methods for assessing the rainfall resilience of transportation networks mostly suffer from the following problems: (1) Simulation methods for modeling rainfall impacts lack realism; (2) After-the-fact evaluations of resilience cannot offer advance warning prior to or during a heavy rain event. To address above problems, we present a novel resilience assessment methodology for evaluating the resilience of road networks in real-time during heavy rainfall scenarios. In this methodology, we propose the temporal decomposition-based dynamic multi-granularity graph neural network (TD2MG2NN) for long-term traffic speed forecasting, providing a perspective on the future evolution of traffic states for accurate resilience assessment. In addition, we construct a composite traffic resilience indicator, designed to comprehensively reflect changes in the spatial-temporal resilience of the transportation system during heavy rain. Experimental results on four publicly real datasets indicate that the prediction performance of TD2MG2NN outperforms state-of-the-art models. The assessment results for the transportation road network in California demonstrate that the comprehensive resilience indicator is superior to single functional resilience indicator and the real-time methodology for evaluating resilience can accurately depict and predict the operation of the road network system under heavy rainfall scenarios.

## 1. Introduction

Assessing how transportation systems handle unusual events has become an increasingly prominent research focus. One significant factor impacting transportation systems is the frequent occurrence of natural disasters such as typhoons, floods and heavy rains. These extreme weather events may result in substantial economic losses, pose a serious risk to public health, and cause damage transportation infrastructure functionality.

Previous assessments of transportation network capabilities primarily focused on indicators such as resistance, vulnerability or robustness, without comprehensively thinking about the ability of urban transportation networks to handle disruptive events. Transportation resilience (or traffic resilience) [1–3] refers to the ability of a transportation system to withstand and recover from disruptive events. The evaluation of traffic resilience is viewed as a new perspective to

measure the performance of transportation systems under disruptive events, and has gained widespread attention among researchers.

Methods for evaluating traffic resilience are rapidly evolving. Currently, a more common paradigm for evaluating the resilience of transportation systems involves the following steps: First, construct resilience assessment indexes based on the structure or functional characteristics, and disturbance event characteristics of transportation systems. Second, monitor changes in resilience indicators before and after disturbance events. Finally, assess the resilience of transportation systems based on these changes. Resilience indicators based on structural features often rely on the topological characteristics of transportation systems. Basic structural resilience metrics include maximum connectivity subgraphs, node strengths, weighted clustering coefficients and average path lengths. In 2015, Testa et al. [4] applied various topological indicators to calculate the impact of extreme weather on

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transportation systems in coastal areas. Resilience indicators based on functional features assess the resilience of transportation systems from in terms of performance, using flow, speed, occupancy rate and other functional values. There is also a relatively specialized category that constructs resilience indicators from the perspective of disturbance event characteristics. For example, Duy et al. [5] constructed FE (flood extent) and VR (vulnerable roads) to assess the characteristics of flood events.

As the intelligent transportation system (ITS) [6,7] continues to develop, spatial-temporal traffic big data lays the foundation for data-driven traffic resilience assessment. Deep learning technology is widely used in traffic data modeling and analysis, which is beneficial for establishing effective emergency preparedness and response procedures. Early researches on traffic data forecasting employs traditional time series methods such as recurrent neural network (RNN) [8], including long short term memory (LSTM) and gated recurrent unit (GRU). Another class of models for traffic data forecasting is convolutional neural network (CNN) and multiple CNN-based models have been proposed, such as WaveNet [9], temporal convolutional network (TCN) [10] and SCINet [11]. Furthermore, beyond time-dependent modeling, recent researches also focus on spatial-temporal dependencies. Graph neural networks (GNNs) are an effective method for capturing the spatial correlations of traffic data. In recent years, many classic GNN-based forecasting models have emerged, including DCRNN [12], GWNet [13] and ASTFGNN [14].

As one of the most common anomalies, the traffic resilience under heavy rainfall scenarios has been widely studied in recent years [15–18]. Our study focuses on the resilience assessment of traffic systems under heavy rain. When heavy rainfall scenario begins, spatial-temporal traffic data may undergo abrupt changes, losing periodic variations in a short period. According to the overview of the resilience assessment of traffic systems under heavy rainfall scenarios, the primary challenges can be succinctly outlined as follows:

- **Lack of real world's traffic data**

Previous work has often modeled the impact of abnormal events on transportation networks based on simulation without the support of real data, and the accuracy warrants further investigation. In the simulation process, many methods make assumptions about complex problems, leading to results that may not match reality.

- **Inaccurate prediction results under heavy rainfall scenarios**

During rainy weather, the traffic network undergoes non-stationary changes in both time and space dimensions, making it challenging to capture temporal periodic features. Several well-known models exhibit poor predictive performance under such conditions. Although aforementioned GNN-based well-known traffic forecast models shown their effectiveness, there are still many key points to emphasize, especially in heavy rainfall conditions:

- How to effectively learn spatial-temporal relations?
- How to correctly capture the non-stationarity of traffic data during heavy rainfall scenarios?
- How to reconcile long and short time period characteristics of traffic data?
- How to reasonably model dynamic spatial dependencies among nodes?

- **Lack of comprehensive resilience indicators**

Recent resilience assessment of traffic networks often rely on static structure or function, unable to holistically reflect the network's resilience. On the one hand, in real scenarios, structure-based indicators almost entirely lack temporal attributes, providing only a static assessment of traffic network resilience against structural damage. On the other hand, though functional resilience indicators provide real-time insights into the operational status of traffic networks, they do not adequately measure the spatial feature of the network.

- **Absence of real-time monitoring and early warning**

Current methods for evaluating resilience primarily focus on modeling and analyzing abnormal attacks in a static environment. This makes it difficult to dynamically track changes in the traffic network and provide early warning notifications to traffic management authorities.

To address the aforementioned issues, this paper proposes a real-time resilience assessment methodology for the road network. The methodology mainly contains four steps, and outputs the resilience assessment result in real time. We propose a deep learning-based model for traffic speed forecasting-temporal decomposition-based dynamic multi-granularity graph neural network (TD2MG2NN) in this methodology. TD2MG2NN improves traffic speed prediction, particularly in heavy rain events. TD2MG2NN includes multiscale temporal decomposition module (MTDM), season-temporal encoder module (STEM), trend-temporal encoder module (TTEM), and dynamic spatial decoder module (DSDM). MTDM decomposes the raw data into seasonal-cycle and trend data through multiscale approach, highlighting different cycle features during heavy rain. STEM based on local-global temporal convolution and TTEM based on regression layer are respectively employed to capture the seasonal and trend features during heavy rain, encoding time series features at different frequencies. Finally, DSDM, which is based on dynamic multi-granularity graph convolution, catches time-varying spatial dependencies, particularly learning the dynamic evolution pattern of traffic speed over time during rainfall events. In addition, proposed methodology constructs a comprehensive spatial-temporal resilience indicator for the road network, integrating speed data and the spatial topological structure. This methodology can dynamically update time-varying resilience assessment value, achieving real-time resilience assessment and early warning for the traffic system. The paper presents a case study on California's traffic network under heavy rainfall scenarios. The contributions of this paper are summarized as follows:

- **Designing a real-time comprehensive resilience assessment methodology**

Overall, the methodology provides real-time resilience results based on different states of the road network at different time slices. Specifically, first, our resilience assessment methodology proposes and applies a deep learning-based model for traffic speed forecasting to study resilience assessment from the perspective of the evolution of the transportation system state, taking into account both past states and future trends. Second, we construct a new resilience assessment indicator to measure the resilience level of road networks from both functional and structural aspects.

- **Proposing a deep learning-based model (TD2MG2NN) for accurate traffic forecasting**

TD2MG2NN is proposed to improve long-term speed forecasting, particularly during heavy rainfall scenarios and thus provides effective data support for resilience assessment. Here are three characteristics in TD2MG2NN:

- An architecture for separating seasonal and trend data based on multiscale time series decomposition. The architecture provides more accurate and explainable results for time series analysis by separating seasonal and trend data in an efficient way.
- Gated local-global temporal convolution for capturing the short and long temporal features. This approach extracts temporal information at different time scales more comprehensively and provides greater flexibility and adaptability in handling time-series data.
- Dynamic multi-granularity graph convolution for extracting multi-level and time-varying spatial dependencies. The framework is capable of adapting to complex spatial structures and is both flexible and robust. This makes it effective

in extracting multilevel and time-varying spatial dependencies, while also providing a more comprehensive perspective for a deeper understanding of spatial relationships and temporal evolution.

- **Constructing a comprehensive spatial–temporal resilience indicator**

A spatial–temporal comprehensive resilience indicator is constructed, integrating both functional and structural features of the urban road network. It can reflect the spatial–temporal changes of the network during and after disruption by heavy rainfall.

## 2. Related work

### 2.1. Resilience assessment

Exploring suitable resilience assessment methods is the primary challenge in resilience research. To precisely quantify the resilience level of traffic networks, evaluation methods can be broadly classified into two categories: topological analysis and data-driven approaches.

Early resilience assessment methods in transportation systems utilized topological approaches. These methods usually simulate disruptive events by systematically removing nodes or links from the traffic network and calculating changes in structural metrics, such as degree, before and after removal. For instance, an example is the work of Testa et al. [4] in 2015, where various topological metrics were applied to assess the impact of extreme weather on coastal traffic systems. More recently, Janic et al. (2020) [19] designed a set of topological metrics based on aviation network attributes to measure network resilience by analyzing numerical changes before and after disturbances and Ma et al. [20] constructed a three-dimensional network structure resilience assessment model based on complex network theory to evaluate the resilience of the transportation system in the Central Shaanxi Plain urban agglomeration in 2022. Topology-based methods have advanced the understanding of traffic resilience by identifying vulnerabilities in the structure of traffic networks under adverse weather conditions. However, they may not accurately reflect the real-world impact of disasters and face challenges in practical applications within transportation systems.

Mainstream evaluation methods of resilience currently utilize a data-driven approach. These methods can be classified into simulation-based data-driven approaches and real-event-based data-driven approaches, depending on the source of disruptive events. Evaluation methods based on data-driven approaches typically involve two steps: First, constructing resilience indicators that incorporate functional data. And second, designing the evaluation methodology to calculate changes in indicators before and after disruptive events. For example, in 2021, Martello et al. [21] examined the impact of sea-level rise on the resilience of coastal urban railway transportation using passenger flow data. Similarly, in 2022, Chen et al. [22] used the system functionality curve, considering the number of operating bus routes and the number of operating taxis as resilience values, to assess the resilience of the road public transportation system in Zhengzhou during the 2021 extreme rainfall disaster. In the same year, Chen et al. [23] simulated short-term operational disturbances and utilized passenger travel time to evaluate the resilience of the Chengdu metro network. Also in 2022, Xu et al. [24] applied historical traffic flow data to assess the resilience of subway networks in five cities including London and Singapore. In 2023, Liu [25] established a new resilience assessment framework, systematically analyzing the coupling mechanism of Beijing's bus-rail transit network, covering both structure and functional characteristics. Again in 2023, Chen [26] chose the expressway network in Fujian Province, China, as the research object and evaluated the resilience of the transportation network under heavy rainfall scenarios. However, these studies, based on traditional statistical models, only provide static case studies for specific real or simulated abnormal events, making it

challenging to draw universal research conclusions. Additionally, the resilience assessment of transportation systems in these studies remains fixed, lacking real-time evolutionary analysis for proactive resilience.

In recent years, deep learning has shown impressive performance in analyzing big data. As intelligent transportation systems continue to develop, real-time traffic data can be collected. The method for evaluating traffic resilience based on deep learning, combined with theories of traffic big data forecast, helps predict specific states and trends of the traffic system during disasters. This facilitates the establishment of efficient emergency preparedness and response procedures by transportation management authorities. In 2020, Wang et al. [27] proposed an end-to-end deep learning framework designed to predict the spatial–temporal patterns of traffic resilience in Shenzhen's road transportation under various extreme weather events, achieving satisfactory predictive results. But this work did not consider the relationship between the resilience assessment itself and traffic state prediction, and uses deep learning models with low accuracy.

### 2.2. Spatial–temporal traffic data forecasting

The traffic forecasting models based on deep learning typically consist of time information mining parts and spatial information mining parts, dividing the extraction of spatial–temporal features into time-dependent modeling and space-dependent modeling.

A major category of deep learning-based models for handling time-dependent relationships is RNN and its variants, such as GRU and LSTM. In 2007, Hutchison et al. introduced FC-LSTM [28], extending RNN to address multi-dimensional problems. In 2015, Shi et al. [29] proposed a convolutional LSTM network, a variant of FC-LSTM. In 2018, Li et al. [12] presented the DCRNN model, employing an “encoder–decoder” structure and utilizing the GRU variant as the time feature extraction module. The AGCRN model proposed by Bai et al. [30] in 2022, Liu et al.'s [31] subway passenger flow prediction PVCGRN model, Li et al.'s [32] DGCRN model in 2021, and Peng Xie's [33] subway passenger flow prediction STDGRL model in 2023 all inherit the research approach of using GRU in the time series processing module from DCRNN. The distinction is made by substituting the linear operation units in GRU with graph convolution modules as proposed by each model. However, RNN-based models may face training difficulties, such as vanishing or exploding gradients, which make them challenging to train when modeling long sequences. Furthermore, RNN-based models may consume excessive resources during time series calculations.

Convolutional neural networks (CNNs) represent another major category of models for extracting time features. Bai et al. [34] introduced one-dimensional causal convolution namely temporal convolutional network (TCN) for modeling time series problems, with training speed and test performance surpassing RNN-based models. Yu et al.'s [35] STGCN model utilized TCN in the time extraction module. Dilated causal convolution is a specific type of one-dimensional convolution that captures longer time dependencies by changing the receptive field size. The GWNet [13] proposed in 2019 and the dynamic hypergraph convolutional Network [36] in 2021 both applied dilated causal convolution in the time series feature extraction module, while ATSGAT [37] in 2022 used multi-channel causal convolution. The MTGNN model proposed by Wu et al. [38] applied parallel operations with dilated causal convolutions of different receptive fields to further expand the receptive field. Although CNN-based methods can capture longer time dependencies by extending the receptive field size, they may struggle to balance dependencies of both long and short duration. An excessively large receptive field may cause the model to overlook short-term time dependencies.

Another deep learning framework for extracting time features is attention mechanisms. Initially applied in the Transformer [39] for natural language processing, attention mechanisms have gradually extended to other domains. The core idea of attention in time series

feature extraction is to capture the relationship between the current traffic state and historical traffic states at the current time point, assigning different weights to historical traffic states when predicting future traffic states. Guo et al. [40] proposed ASTGCN, which initially combined attention mechanisms with convolutional networks and applied them to the time series feature processing module of traffic prediction. Zheng et al.'s [41] GMAN model added node embedding and transition attention on the basis of attention mechanisms. Guo et al. [14] introduced the ASTGNN model for traffic data prediction, based on the Transformer structure, demonstrating the continued superiority of attention-based Transformers in the field of traffic forecast. In 2022, Shao et al. [42] proposed the D2STGNN model, which utilized the GRU variant for extracting short-term dependencies and employed multi-head attention mechanisms to capture long-term dependencies. In 2023, Nie et al. [43] proposed segmenting time series into subsequence-level fragments input to the Transformer, improving the predictive accuracy of attention mechanisms in time series prediction.

Traditional convolutional operations can only handle grid structures and are unable to address non-Euclidean structures, such as spatial traffic data. In recent years, several models based on graph neural network (GNN) have demonstrated superior performance in traffic prediction compared to previous methods. In 2014, Bruna et al. [44] first proposed spectral graph convolution, involving the computation of the Laplacian matrix's eigenvalue decomposition, converting [45] the graph data into the frequency domain by a Fourier transform, conducting convolution operations in the frequency domain, and finally transforming the frequency domain data back to the spatial domain. Due to the high computational complexity of eigenvalue decomposition, the initial spectral domain convolution methods had high training requirements and limited applicability. ChebNet introduced Chebyshev polynomials of the eigenvalue diagonal matrix as convolution kernels, reducing the complexity of spectral domain convolution algorithms. The further first-order ChebNet [46] simplified the convolution operation without sacrificing performance. The STGCN model [35] applied ChebNet for traffic forecasting. Spatial graph convolution methods directly define graph convolution as aggregating information from central nodes and their neighboring nodes to update the central node's information.

The DCRNN model [12] initially modeled the traffic network as a directed graph, representing the transmission of traffic states between traffic nodes as a diffusion process, and introduced diffusion graph convolution. Building upon the diffusion graph concept of DCRNN, GWNNet [13] further proposed an adaptive state transition matrix, enabling the model to learn hidden spatial information beyond predefined graphs. The DGCNN model in 2023 [32], the D2STGNN model in 2022 [42], and others adopted diffusion graph convolution to update traffic node information. The MTGNN model proposed by Wu et al. [38] introduced mixed skip-mode diffusion graph convolution, allowing central nodes to integrate information from neighboring nodes. Apart from GNN, there are spatial dependency modeling methods based on graph attention mechanisms. Similar to the idea of temporal attention mechanisms, the traffic state of one network node is influenced by the traffic states monitored at other network nodes, with different weights assigned to the traffic states of other location nodes when predicting. Graph attention convolution network (2020) [47] and PDFormer (2023) [48] both applied different attention mechanism methods to traffic prediction. MegaCRN [49] in 2023 proposed a novel graph learning framework called spatial-temporal meta graph learning. However, some studies [50] have pointed out that GNN layers can lead to over-smoothed graph data. That is, information among becomes increasingly similar, ultimately resulting in reduced distinctiveness between nodes. This is because each layer's graph convolution operation averages the embeddings of nodes (a unique aggregation operation of GCN), leading to information blending and smoothing. As a result, many studies have begun to explore abandoning graph neural networks and researching

spatial-temporal traffic forecast based on the inseparability of spatial-temporal data. In 2021, ST-Norm [51] proposed two normalization modules—time normalization and space normalization to refine high-frequency components and local components in the original data. In 2022, STID [52] achieved satisfactory predictive results solely through multilayer perceptron (MLP) model by appending spatial and temporal identity information. However, while achieving better results in regular scenarios, these models are relatively ineffective in heavy rainfall scenarios. This limitation undermines their usefulness in resilience assessment due to the complex temporal instability and spatial evolutionary relationships of traffic states.

### 3. Preliminaries and problem statement

#### 3.1. Preliminaries

**Definition 1 (Traffic Network).** We define the topological graph of a road network (directed or undirected) as  $G = (V, E)$ , where  $V$  is a set of  $N$  nodes,  $|V| = N$ ,  $|E| = M$  and each node represents a traffic data monitoring point.  $E$  is the set of  $M$  edges among nodes.

**Definition 2 (Traffic State Matrix).** At time slice  $t$ , the observed traffic state of the traffic network  $G$  is represented by the traffic state matrix  $X_t$ , where  $X_t = (X_{t,0}, X_{t,1}, X_{t,2}, \dots, X_{t,N-1})^T$  belongs to  $R^{N \times C}$  and  $C$  is the number of traffic states. Here,  $X_{t,v}$  belongs to  $R^C$ , indicating the traffic state vector of node  $v$  at time slice  $t$ .

#### 3.2. Problem statement

In this paper, the problem is traffic resilience assessment under the heavy rain scenarios. At time slice  $t$ , we quantify the evaluation results of the traffic system's ability to resist the impacts of heavy rainfall and recover its traffic functions based on a traffic resilience indicator over a historical time window. At time  $t$ , the definition of the sequence of recent historical spatial-temporal traffic state matrices, denoted as  $H = (X_{t-T_{his}+1}, X_{t-T_{his}+2}, \dots, X_t)$ ,  $\in R^{N \times C \times T_{his}}$ , covering the past  $T_{his}$  time slices. The resilience assessment value of the traffic system at rainy time  $t$  is defined as  $RV(t)_{eval, graph}$ , obtained by a assessment methodology using the historical traffic states over a window of  $T_{his}$  time slices and the traffic network topology  $G$ , expressed as follows in Eq. (1),

$$RV(t)_{eval, graph} = Eval(H, G) \quad (1)$$

where  $Eval$  represents the traffic resilience assessment methodology. Our innovative methodology is based on traffic forecasting to analyze the resilience's evolutionary pattern. At time  $t$ ,  $H$  is used to forecast the sequence of traffic state matrices  $Y = (X_{t+1}, X_{t+2}, \dots, X_{t+T_{fut}})$ , belonging to  $R^{N \times C \times T_{fut}}$ , within the future  $T_{fut}$  time slices. The core of traffic forecasting lies in the mapping relationship from historical traffic state matrices to future traffic state matrices, expressed as following in Eq. (2),

$$H \xrightarrow{g} Y \quad (2)$$

where  $g$  represents the mapping relationship from historical traffic state matrices to future traffic state matrices.

### 4. Methodology

#### 4.1. Real-time resilience assessment

Fig. 1 visualizes the overall process of real-time comprehensive resilience assessment methodology. The input for the methodology is the speed matrix of the historical time slices of the road network at the time slice  $t$  during the heavy rain event to be evaluated. The output is the resilience assessment value of the road network at time slice  $t$ . The method predicts the impact of rainfall on the road network in advance



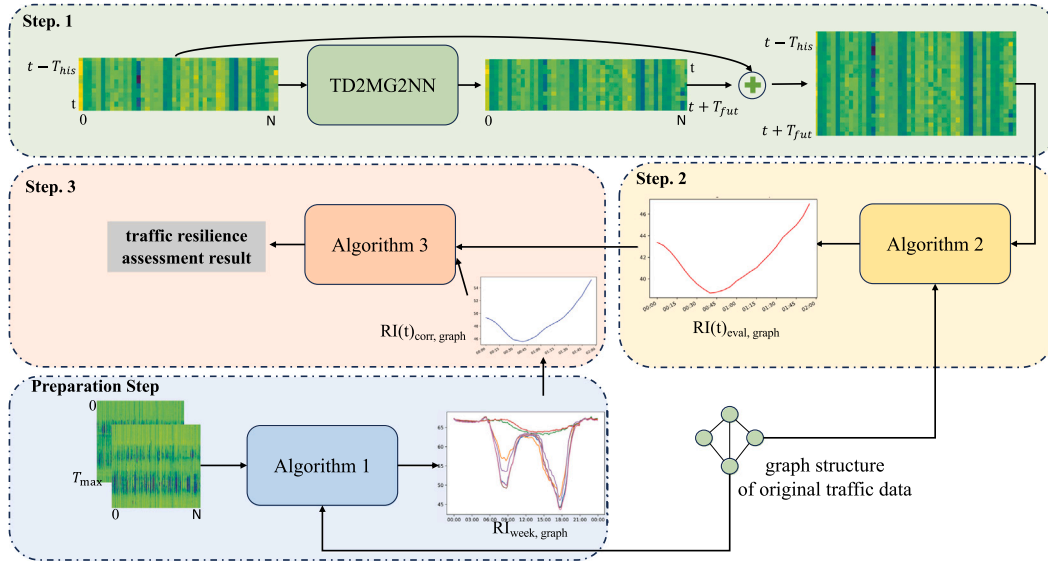


Fig. 1. Workflow of the real-time comprehensive resilience assessment methodology under heavy rainfall scenarios.

to achieve predictive resilience assessment. The methodology consists of four steps, Preparation Step, Step.1, Step.2 and Step.3. In Preparation Step, the input comprises all historical speed matrices of the training set and the road network structure. The output of Preparation Step, processed by Algorithm 1, is the benchmark of comprehensive resilience assessment indicators under normal weather conditions (one week). Then in Step.1, the input is the speed matrix of the historical time slices of the road network at time slice  $t$  to be assessed. Through the proposed speed forecasting model temporal decomposition-based dynamic multi-granularity graph neural network (TD2MG2NN, see Fig. 2), the algorithm generates the speed matrix for the predicted future time slices at  $t$ . The output is the concatenated speed matrix of historical and predicted values. Step.2's input is the output from Step.1. The input  $RI(t)_{eval,graph}$  is processed by Algorithm 2, which outputs the time-varying curve of comprehensive resilience indicator within this time period. Finally, Step.3 takes the output of Step.2,  $RI(t)_{eval,graph}$ , time point at which the rain started,  $t_{begin}$ , the comprehensive resilience indicator from the onset of rainfall to the commencement of assessment,  $RI'(t)_{eval,graph}$ , and the benchmark comprehensive resilience indicator,  $RI_{week,graph}$ . Step.3 evaluates the resilience of the road network from the time point when resilience capability recovers to 95% of normal level to the start of rainfall, assessing the network's resilience during this period. The output is the final resilience assessment result,  $RV(t)_{eval,graph}$ , processed by Algorithm 3. In Section 4.3, we provide the formal definitions and computational formulas for  $RI$ ,  $RC$ , and  $RV$ .

#### 4.2. Traffic speed forecasting model

##### 4.2.1. Multiscale temporal decomposition module

The concept of multiscale temporal decomposition involves creating a multiscale pooling layer that utilizes various kernels of different sizes to differentiate between season-cycle and trend data with distinct patterns. This approach enhances the model's ability to extract temporal features by effectively adjusting to periodic features of varying scales in the data. The use of different pooling kernel sizes enables the extraction of diverse periodic features from the original time series data. By considering multiple scales, the model can comprehensively capture the complexity and variations in the data. Eqs. (3) and (4) illustrate the operation of decomposing season-cycle data and trend data.

$$H_s = 1/s \sum_{i=0}^{s-1} AvgPool(H)_{kernel_i} \quad (3)$$

#### Algorithm 1: Computing Benchmark of Comprehensive Resilience Indicators

```

1 // D is the number of historical days;
2 // I is the number of sensors;
3 // T is the number of samples within one day;
inputs:  $H_s \in R^{D \times I \times T}$ , historical speed matrices of the road network
          $Wet_{day} \in R^{D \times 24}$ , 24-hour weather
output:  $RI_{week,graph} \in R^{7 \times T}$ , benchmarks of comprehensive resilience assessment indicators

4  $D_{norm} \leftarrow \{\}$ 
5 for the  $d_{th}$  day in  $D$  days do
6   if  $Wet_{day}^d < 20mm$  then
7      $D_{norm} \leftarrow D_{norm} \cup \{d\}$ ;
8   end
9 end
10 for the  $d_{th}$  day in  $D$  days do
11   for the  $t_{th}$  slice in  $T$  slices do
12     for the  $i_{th}$  sensors in  $I$  sensors do
13       compute  $RI_{day,node}^{d,t,i} \in R^1$  through the method for comprehensive resilience indicator of single node;
14     end
15     compute  $RI_{day,graph}^{d,t} \in R^1$  through the method for comprehensive resilience indicator of the graph;
16   end
17 end
18 for the  $j_{th}$  day in a week  $J = \{0, 1, 2, 3, 4, 5, 6\}$  do
19   for the  $t_{th}$  slice in  $T$  slices do
20     compute  $RI_{week,graph}^{j,t} \in R^1$ , resilience indicators in a week according to  $D_{norm}$ ;
21   end
22 end

```

$$H_t = H - H_s \quad (4)$$

$H$  represents the original traffic speed data,  $H_s$  and  $H_t$  are the decomposed season-cycle and trend data,  $AvgPool$  denotes the average pooling operation,  $kernel_i$  refers to the  $i$ th convolutional kernel, and  $s$

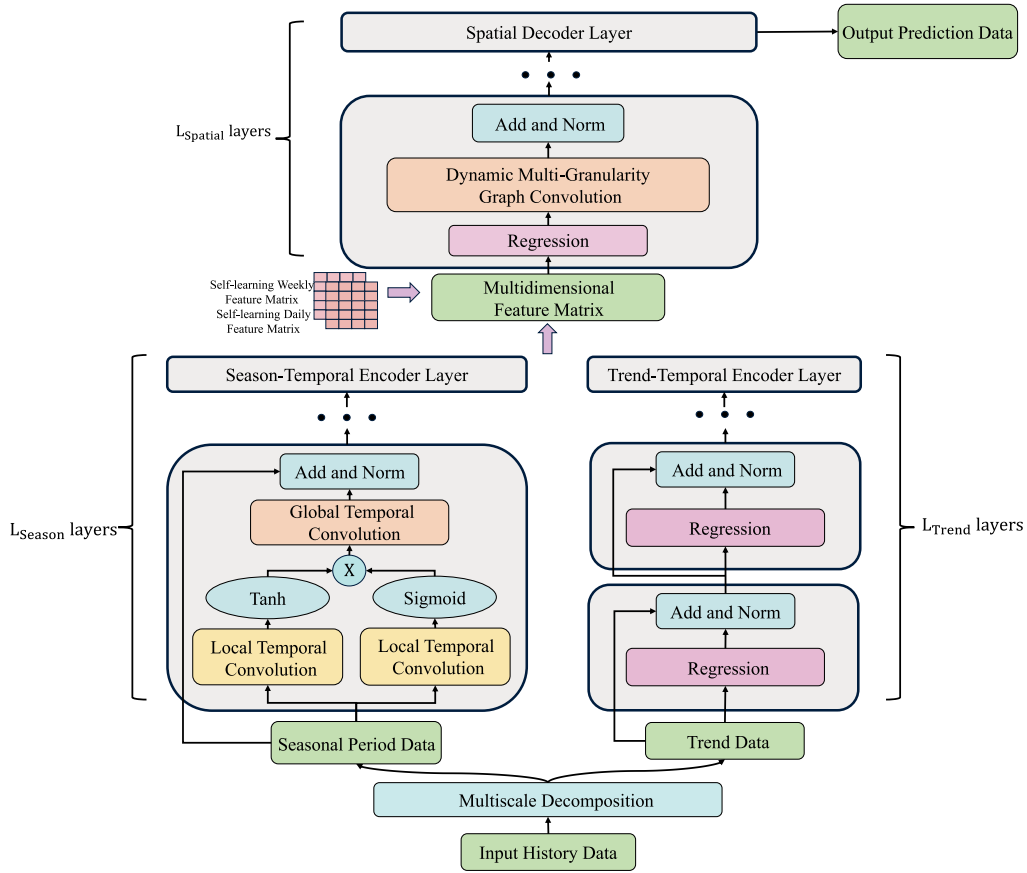


Fig. 2. The architecture of temporal decomposition-based dynamic multi-granularity graph neural network (TD2MG2NN).

**Algorithm 2:** Computing Comprehensive Resilience Indicators During the Heavy Rain Event

```

1 //  $T_{his}$  is the length of historical time slice;
2 //  $T_{fut}$  is the length of predicted time slice;
3 //  $I$  is the number of sensors;
   inputs:  $Z_t \in R^{I \times (T_{fut} + T_{his})}$ , the result of concatenating historical
           speed values  $X_t$  and predicted speed values  $Y_t$ 
   output:  $RI(t)_{eval, graph} \in R^{(T_{his} + T_{fut})}$ , the time-varying
           comprehensive resilience indexes

4 for the  $k_{th}$  time slice in  $(T_{his} + T_{fut})$  slices do
5   for the  $i_{th}$  sensor in the road network do
6     compute  $RI_{eval, node}^{j,i} \in R^1$ , through the method for
           comprehensive resilience indicator of single node;
7   end
8   compute  $RI_{eval, graph}^j \in R^1$ , through the method for
           comprehensive resilience index of the graph;
9 end
10  $RI(t)_{eval, graph} \leftarrow \{RI_{eval, graph}^{1-T_{his}}, \dots, RI_{eval, graph}^{T_{his}+T_{fut}}\};$ 

```

is the number of pooling kernels, the seasonal cycle data  $H_s$  is the result of multiple average pooling operations on the original time series data, while the trend data is obtained by subtracting the seasonal cycle data from the original time series data.

#### 4.2.2. Trend-temporal encoder module

Trend data refers to the overall direction or tendency of raw time series data over an extended period. It indicates a gradual increase

or decrease in data over time with a clear direction, which can be ascending, descending, or maintaining stability. Trend data typically reflects the fundamental changing trends in a system or process. During periods of heavy rainfall, the instability of trend data intensifies. The model deals with trend data by utilizing multiple layers of trend encoding layers to handle the rapid changes in trend-cycle data. This approach captures the unstable temporal features in trend-cycle data during heavy rain. The regression layer of the trend encoding layer uses linear regression methods and residual connections, which facilitate more effective gradient propagation, making it easier for the network to learn residuals. The processing steps mentioned above are shown in Eqs. (5), (6), and (7),

$$F_t^0 = H_t \quad (5)$$

$$F_t^l = BN(ReLU(W_t^l \times F_t^{l-1} + b_t^l) + F_t^{l-1}) \quad (6)$$

$$M_t = F_t^{L_t} \quad (7)$$

$H_t$  represents the trend-cycle data,  $F_t^l$  denotes the output of the  $l$ th layer of the trend-cycle encoding layer,  $W_t^l$  is the weight matrix of the  $l$ th layer linear regression,  $b_t^l$  is the bias of the  $l$ th layer linear regression,  $L_t$  is the number of layers in the trend-cycle encoding layer, and  $M_t$  is the resulting matrix of trend-cycle feature encoding.  $BN$  refers to the batch normalization operation, which normalizes the input of each layer in the network, allowing it to be scaled and shifted through learnable parameters. This process maintains a zero mean and unit variance for the data during the training process.  $ReLU(*)$  represents a non-linear activation function.

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**Algorithm 3:** Computing Resilience Assessment Result During the Heavy Rain Event
 

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**inputs:**  $t_{begin}$ , the most recent point at which it started raining,  
 $RI'(t)_{eval, graph} \in R^{t-T_{his}-t_{begin}}$   
 $RI(t)_{eval, graph} \in R^{(T_{his}+T_{fut})}$   
 $RI_{week, graph} \in R^{7 \times T}$   
**output:**  $RV(t)_{eval, graph} \in R^1$ , the resilience assessment result at time slice  $t$

```

1  $t_{end} \leftarrow -1$ 
2 for the  $k_{th}$  time slice in  $t - T_{his} - t_{begin}$  slices do
3   compute  $RC_{eval, graph}^{t, k} \in R^1$ , which is the ratio of
      $RI'(t)_{eval, graph}$  to  $RI_{week, graph}$  during the corresponding day
     and time slices;
4 end
5 for the  $k_{th}$  time slice  $T^k$  in  $(T_{his} + T_{fut})$  slices do
6   compute  $RC_{eval, graph}^{t, k} \in R^1$ , which is the ratio of
      $RI(t)_{eval, graph}$  to  $RI_{week, graph}$  during the corresponding day
     and time slices;
7   if  $RC_{eval, graph}^{t, k} \geq 0.95$  then
8      $t_{end} \leftarrow T^k$ 
9     break
10  end
11 end
12 if  $t_{end} \geq 0$  then
13    $RV(t)_{eval, graph} \leftarrow$ 
      $(\int_{t_{begin}}^{t-T_{his}} RC_{eval, graph}^{t, k} dk + \int_{t-T_{his}}^{t_{end}} RC_{eval, graph}^{t, k} dk) / \int_{t_{begin}}^{t_{end}} 1 dk;$ 
14 end
  
```

---

#### 4.2.3. Season-temporal encoder module

Seasonal cycles represent recurring patterns in traffic data over short periods, often arising from regular factors such as traffic signals. Seasonal cyclic effects typically cycle within relatively short time frames. Previous one-dimensional temporal convolutions lack an explicit mechanism for modeling long-term dependencies. During heavy rain, the impact of precipitation on traffic conditions can alter the temporal dependencies of seasonal cycles over a certain period, making it challenging to capture global temporal features. Although stacking multiple convolutional layers in one-dimensional temporal convolutions can capture dependencies within a certain range, limitations may still exist for longer time spans. This paper proposes a gated one-dimensional local-global temporal convolution module. It utilizes gating mechanisms to control the output of local temporal features and further extracts global temporal features through global temporal convolutions. This approach enhances the model's representation ability for global temporal features during heavy rain. The aforementioned processing steps are illustrated in Eqs. (8) to (12),

$$F_s^0 = H_s \quad (8)$$

$$F_{s_{lo}} = \text{Tan}(\text{Conv1D}_{lo}(F_s^{l-1})) \odot \text{Sig}(\text{Conv1D}'_{lo}(F_s^{l-1})) \quad (9)$$

$$F_{s_{gl}}^l = \text{ReLU}(\text{Conv1D}_{gl}(\text{Padding}(F_{s_{lo}}^l))) \quad (10)$$

$$F_s^l = \text{BN}(F_s^{l-1} + F_{s_{gl}}^l) \quad (11)$$

$$M_s = F_s^{L_{season}} \quad (12)$$

$H_s$  represents season-cycle data,  $F_s^l$  denotes the output of the  $l$ th layer of the season-temporal encoding layer,  $F_{s_{gl}}^l$  is the local seasonal cycle feature matrix for the  $l$ th layer, and  $F_{s_{gl}}^l$  is the global seasonal cycle feature matrix for the  $l$ th layer.  $L_{season}$  represents the number of layers in the seasonal cycle data encoding layer.  $\text{Tan}$  is the tanh activation

function,  $\text{Sig}$  is the sigmoid activation function, and  $\odot$  represents the element-wise multiplication operation.  $\text{Conv1D}_{lo}$  and  $\text{Conv1D}'_{local}$  are two different one-dimensional local temporal convolutions,  $\text{Conv1D}_{gl}$  is the one-dimensional global temporal convolution, and  $\text{Padding}$  is used to zero-pad the temporal data by  $\text{len}(F_{s_{lo}}^l)$  time steps forward, ensuring that the temporal span of  $F_{s_{lo}}^l$  remains unchanged before and after one-dimensional convolution. The convolutional kernel size of  $\text{Conv1D}_{gl}$  is  $(1, \text{len}(F_{s_{lo}}^l) + 1)$ , where  $\text{len}(F_{s_{lo}}^l)$  is the temporal span of  $F_{s_{lo}}^l$ .  $M_s$  represents the final matrix of seasonal cycle feature encoding.

In the context of heavy rainfall scenarios, traffic data may exhibit certain hidden features that cannot be captured by trend-cycle and seasonal cycle patterns. This paper enhances the model's ability to handle details in traffic data during heavy rainfall scenarios by constructing a self-learning time feature encoding matrix. The process of constructing self-learning daily and weekly periodic feature encoding matrices and concatenating these temporal feature encoding matrices is illustrated in Eqs. (13) to (15),

$$M_d = E_d[H_{time}, :] \quad (13)$$

$$M_w = E_w[H_{day}, :] \quad (14)$$

$$M_{feature} = \text{Cat}(M_s, M_t, M_d, M_w) \quad (15)$$

$E_d \in R^{N_{time} \times N_{daily}}$  and  $E_w \in R^{N_{day} \times N_{weekly}}$  respectively represent the self-learning embedding matrices for daily and weekly periodic features.  $N_{time}$  represents the frequency of traffic sampling conducted within a 24-h period.  $N_{daily}$  is the feature dimension of daily periodic features,  $N_{day}$  is the number of days in a week (default is 7 days), and  $N_{weekly}$  is the feature dimension of weekly periodic features.  $H_{time}$  and  $H_{day}$  are the sampling time points within a day and sampling day point within a week for the input spatial-temporal traffic data, respectively.  $M_d$  is the selected feature encoding matrix corresponding to the sampling time points of the input data,  $M_w$  is the selected feature matrix corresponding to the sampling days of the input data, and  $M_{feature}$  is the final multidimensional time feature encoding matrix obtained.

#### 4.2.4. Dynamic spatial decoder module

For spatial dependencies modeling, dynamic multi-granularity graphs perceive and aggregate time-varying graph structural information at different spatial scales. The concept of edges is extended to hyperedges, which refer to different ways of partitioning a graph into hyperedges at distinct granularity levels. A graph with a conventional structure represents the most specific granularity, with the number of hyperedges being equal to the number of nodes. Hypergraph-level granularity is employed to learn higher-order relationships in traffic speed between regions that evolve over time. Various hyperedge partitioning methods correspond to different regional partitioning rules, as exemplified in Fig. 3. Fig. 4 shows that the proposed model uses dynamic multi-granularity graph convolution in a parallel manner to model spatial dependencies across various granularity levels. Eqs. (16) through (21) and Eqs. (22) through (25) respectively describe the computation process for the dynamic node-level and hyperedge-level graph convolution operations. Fig. 5 describes the process of computing the node-node incidence matrix and Fig. 6 shows how to compute the node-hyperedge incidence matrix.

$$T_f^1 = D^{-\frac{1}{2}} A D^{-\frac{1}{2}} \quad (16)$$

$$T_b^1 = D^{\text{transpose} - \frac{1}{2}} A D^{\text{transpose} - \frac{1}{2}} \quad (17)$$

$$T A_{node} = A_{node} \| T_{day} \| T_{week} \quad (18)$$

$$T_{node}^1 = (T A_{node})(T A_{node}^{\text{transpose}}) \quad (19)$$

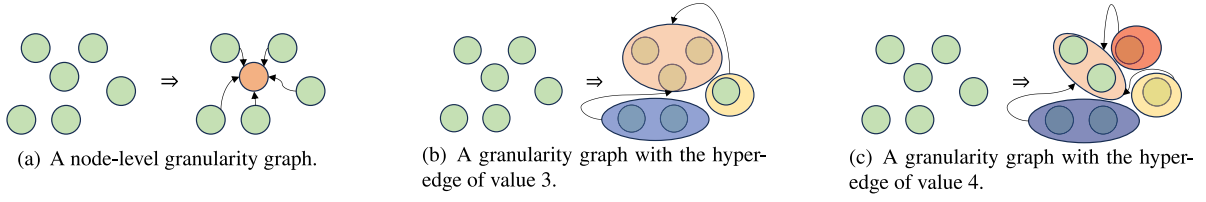


Fig. 3. An example of multi-granularity graphs. This diagram illustrates the information aggregation of node-level granularity graphs, granularity graphs with hyperedges of 3 and 4.

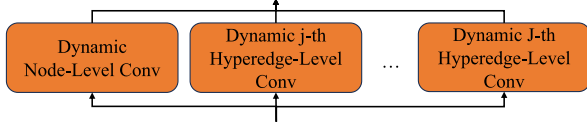


Fig. 4. The structure of dynamic multi-granularity graph convolution.

$$F_G^0 = \text{Linear}(M_{\text{feature}}) \quad (20)$$

$$F_{\text{node}}^l = \sum_{i=1}^{P_0} (T_f^i F_G^{l-1} W^{l_0} + T_b^i F_G^{l-1} W^{l_1} + T_{\text{node}}^i F_G^{l-1} W^{l_2}) \quad (21)$$

$$TA_{\text{hyper}_j} = A_{\text{hyper}_j} \|T'_{\text{day}}\|T'_{\text{week}} \quad (22)$$

$$T_{\text{hyper}_j}^1 = TA_{\text{hyper}_j} TA_{\text{hyper}_j}^{\text{transpose}} \quad (23)$$

$$F_{\text{hyper}_j}^l = \sum_{i=1}^{P_1} (F_G^{l-1} W_{\text{hyper}_j}^{l_3} (F_G^{l-1} (F_G^{l-1} W_{\text{hyper}_j}^{l_3})^{\text{transpose}} F_G^{l-1})) \quad (24)$$

$$F_G^l = \text{BN}(F_{\text{node}}^l + \sum_{j=1}^J (F_{\text{hyper}_j}^l) + F_G^{l-1}) \quad (25)$$

$$Y_t = F_G^{L_{\text{spatial}}} \quad (26)$$

Eqs. (16) and (17) compute the first-order forward and backward node-node incidence matrices  $T_f^1$  and  $T_b^1$ , where  $D$  is the graph's degree matrix,  $A$  is the adjacency matrix and  $D^{\text{transpose}}$  is the transpose of  $D$ . Eqs. (18) to (19) calculate the first-order dynamic node-node incidence matrix  $T_{\text{node}}^1$ , where  $A_{\text{node}} \in R^{N_{\text{node}} \times N_{\text{node}}}$  is the self-learning spatial embedding matrix,  $N_{\text{node}}$  is the embedding dimension.  $T_{\text{day}} \in R^{N_{\text{day}} \times d_{\text{day}}}$  and  $T_{\text{week}} \in R^{N_{\text{week}} \times d_{\text{week}}}$  are self-learning dynamic daily and weekly embedding matrices,  $d_{\text{day}}$  and  $d_{\text{week}}$  are corresponding embedding dimensions.  $TA_{\text{node}}$  denotes the dynamic node-node incidence matrix and  $TA_{\text{node}}^{\text{transpose}}$  is the transpose of  $TA_{\text{node}}$ .  $F_G^0$  represents the output of the 0-th layer, whose value is the result of a linear map of  $M_{\text{feature}}$ .  $F_{\text{node}}^l$  is the output of dynamic node-level granularity graph convolution in the  $l$ -th layer and  $P_0$  represents the maximum order,  $T_f^l$  and  $T_b^l$  are the  $l$ -th order forward and backward node-node incidence matrices and  $T_{\text{node}}^l$  is the  $l$ -th order dynamic node-node incidence matrix.  $W^{l_0}$ ,  $W^{l_1}$  and  $W^{l_2}$  are three different weight matrices. Eqs. (22) to (23) calculate the first-order dynamic  $j$ -th hyperedge-hyperedge incidence matrix  $T_{\text{hyper}_j}^1$ , resembling the construction process of node-node incidence matrix. In Eq. (24),  $F_{\text{hyper}_j}^l$  denotes the output of the dynamic  $j$ -th hyperedge-level granularity graph convolution in the  $l$ -th layer,  $P_1$  represents the maximum order and  $W_{\text{hyper}_j}^{l_3}$  is the self-learning node-hyperedge incidence matrix. In Eq. (25),  $F_G^l$  represents the output of the  $l$ -th layer and  $J$  denotes the number of the granularity. In Eq. (26),  $Y_t \in R^{N \times V}$  represents the final output of the traffic speed prediction, where  $V$  represents the length of time slices.

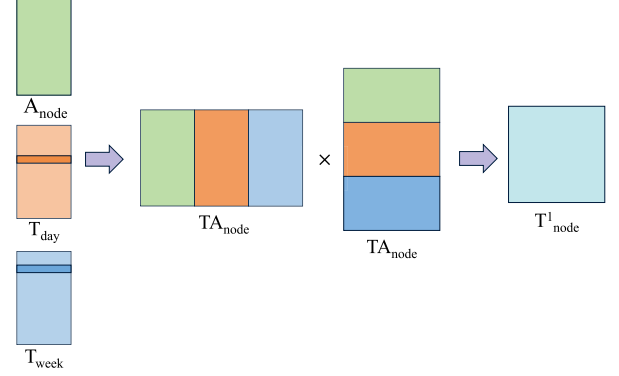


Fig. 5. The process of computing the dynamic node-node incidence matrix.

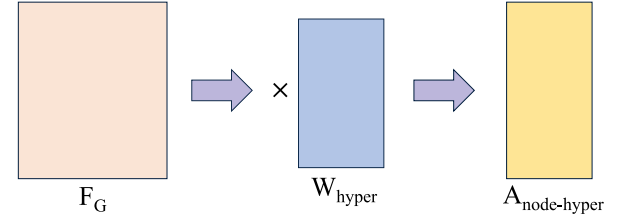


Fig. 6. The process of computing the dynamic node-hyperedge incidence matrix.

#### 4.3. Calculating resilience indicator, capability, and assessment value

The current mainstream's single resilience indicator based on either structure or function often fails to comprehensively reflect the overall resilience of the road network. For example, structure-based resilience indicators can only statically reflect the structural damage caused by exceptional events to the road network. However, the impact of common anomalous events such as heavy rainfall on the structural damage of the road network is not obvious. Structure-based resilience indicators are often considered to be time-invariant under heavy rainfall scenarios, which makes it difficult to quantitatively assess the resilience of the road network. On the other hand, function-based resilience indicators exhibit time-varying characteristics and can intuitively reflect the real-time operational status of the road network. However, they may overlook the impact of certain critical nodes, such as traffic hubs on the overall road network. This paper proposes a comprehensive algorithm for constructing resilience indicators from both structural and functional perspectives, addressing the aforementioned issues.

The structural indicators of nodes employed in this study encompass three categories: node degree centrality, node betweenness centrality, and node reachability. Specifically, node degree centrality is defined as the degree of a node divided by the maximum possible degree. Node betweenness centrality is calculated as the sum of the occurrences of a node in all shortest paths divided by the total number of shortest paths. Node reachability refers to the number of nodes that a given node can reach within a fixed step length. The computation methods for these



three indicators are illustrated by Eqs. (27) to (29),

$$C_D(v) = \frac{\deg(v)}{N-1} \quad (27)$$

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma} \quad (28)$$

$$R_E(v) = \sum_{k=1}^g \frac{n_k(v)}{N} \quad (29)$$

$N$  represents the number of nodes in the graph,  $v$ ,  $t$  and  $s$  are denoted as node;  $C_D(v)$  is node degree centrality of  $v$ ;  $\deg(v)$  represents the degree of  $v$ ;  $C_B(v)$  is node betweenness centrality of  $v$ , the total number of shortest paths from node  $s$  to node  $t$  is denoted as  $\sigma_{st}(v)$ ;  $R_E(v)$  is node reachability of  $v$ ,  $g$  represented a fixed max stride,  $n_k(v)$  is the number of nodes that a node can reach at stride  $k$ .

Subsequently, the comprehensive resilience indicator of node is calculated by combining node structure and functional metrics. The calculation method for node structure resilience indicator is provided in Eq. (30), the calculation method for node functional resilience indicator is given in (31), and the calculation method for the comprehensive node resilience indicator is presented in Eq. (32),

$$RI_{str,node}^v = w_0 * C_D(v) + w_1 * C_B(v) + w_2 * R_E(v) \quad (30)$$

$$RI_{func,node}^v = Speed(v) \quad (31)$$

$$RI_{node}^v = RI_{str,node}^v * RI_{func,node}^v \quad (32)$$

$Speed(v)$  represents the speed value of node  $v$  at a specific time slice.  $w_0$ ,  $w_1$  and  $w_2$  are weight coefficients.  $RI_{str,node}^v$  denotes the structural resilience indicator of node  $v$ , while  $RI_{func,node}^v$  signifies the functional resilience indicator of node  $v$ .  $RI_{node}^v$  is the comprehensive resilience indicator of node  $v$ , calculated as the product of the structural and functional resilience indicator. The calculation method for the comprehensive resilience indicator (RI) of the road network is given in the formula (33),

$$RI = \sum_v RI_{node}^v \quad (33)$$

$RI$  represents the comprehensive resilience indicator of the urban transportation network, which is the sum of the comprehensive resilience indicators for all nodes in the network. The specific calculation process can be found in Algorithm 1 and Algorithm 2. The comprehensive resilience capability ( $RC$ ) is the ratio of  $RI$  during the heavy rainfall phase to  $RI$  under normal weather conditions, as formally defined in Eq. (34).  $RI_{eval}$  denotes the resilience indicator at a specific time during the heavy rainfall period, while  $RI_{norm}$  represents the resilience indicator at the corresponding time under normal weather conditions. The corresponding time is defined as the same day of the week and the same time point within the day.  $RC$  can clearly reflect the degree of resilience indicators degradation caused by rainfall.

$$RC = \frac{RI_{eval}}{RI_{norm}} \quad (34)$$

Furthermore, the resilience assessment value ( $RV$ ) is defined as the ratio of the integral of  $RC$  over the period from the onset of heavy rainfall until its impact completely dissipates, to the integral of the same period when  $RC$  equals 1, in Eq. (35).  $RV$  is utilized to assess the overall resilience of the traffic system during heavy rainfall. We use  $t_0$  to denote the onset time of the rainfall event and  $t_1$  to denote the time when the rainfall impact dissipates. Detailed calculation procedures for  $RC$  and  $RV$  provided in Algorithm 3.

$$RV = \frac{\int_{t_0}^{t_1} RC dk}{\int_{t_0}^{t_1} 1 dk} \quad (35)$$

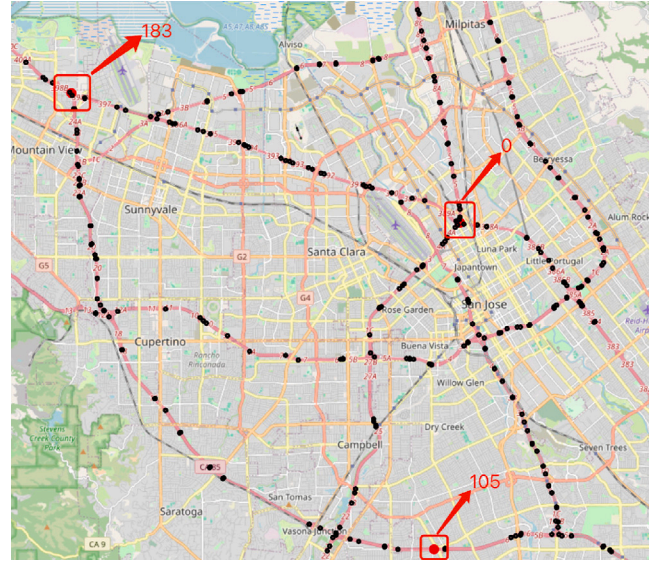


Fig. 7. Spatial distribution of sampling nodes of PeMS-Rain in the San Jose area.

Table 1

Statistics of public traffic speed datasets.

| Dataset   | Nodes | Length  | Sample rate | Time span       |
|-----------|-------|---------|-------------|-----------------|
| PeMS-BAY  | 325   | 52,116  | 5 min       | 01/2017–05/2017 |
| PeMSD7(M) | 228   | 12,672  | 5 min       | 05/2012–06/2012 |
| PeMSD7(L) | 1026  | 12,672  | 5 min       | 05/2012–06/2012 |
| PeMS-Rain | 308   | 105,120 | 5 min       | 03/2018–02/2019 |

## 5. Experiments

### 5.1. Experimental setup

#### 5.1.1. Datasets

The experiments utilize four real-world datasets, which can be categorized into two main types: first, public traffic speed datasets, including PeMS-BAY, PeMSD7(M) and PeMSD7(L); second, a self-building traffic speed dataset with rainfall characteristics, denoted as PeMS-Rain and Fig. 7 illustrates the spatial distribution of sampling nodes of PeMS-Rain. Like many researches, we conduct experiments on three commonly used spatial-temporal sequence datasets, all derived from the Californian Performance of Transportation system. Additionally, due to the limited coverage of intense rainfall periods in public datasets, for a more comprehensive investigation into the resilience assessment of transportation systems under heavy rain, we opt for the road network transportation system in the San Jose area (spatial range: 121°75'W ~ 122°07'W, 37°23'N ~ 37°45'N). We build a novel spatial-temporal speed dataset for this specific region, referred as PeMS-Rain, and record data for all days with 20 mm or more of rainfall and label them as heavy rain events. Detailed information is provided in Table 1.

#### 5.1.2. Baselines for traffic speed forecasting

We compare TD2MG2NN against the following ten well-known baseline models to demonstrate its superiority in traffic speed forecasting under both normal traffic conditions and heavy rainfall periods.

- DCRNN [12]: DCRNN is the diffusion convolutional recurrent neural network, which utilizes diffusion graph convolution and GRU to capture temporal and spatial dependencies, respectively.
- STGCN [35]: STGCN is the spatial-temporal graph convolutional network, which employs conventional graph convolution and one-dimensional convolution to capture temporal and spatial dependencies, respectively.

**Table 2**

Prediction results of TD2MG2NN and other state-of-the-art baselines on PeMS-BAY, PeMSD7(M) and PeMSD7(L). The best results are highlighted in bold, while the second-best results are underscored.

| Model    | Metrics | PeMS-BAY |       |       |       | PeMSD7(M) |       |        |        | PeMSD7(L) |       |        |       |
|----------|---------|----------|-------|-------|-------|-----------|-------|--------|--------|-----------|-------|--------|-------|
|          |         | @3       | @6    | @12   | avg   | @3        | @6    | @12    | avg    | @3        | @6    | @12    | avg   |
| DCRNN    | MAE     | 1.31     | 1.67  | 1.99  | 1.62  | 2.62      | 3.56  | 5.03   | 3.58   | 2.79      | 3.78  | 5.27   | 3.79  |
|          | RMSE    | 2.80     | 3.81  | 4.66  | 3.74  | 4.99      | 6.85  | 9.29   | 7.02   | 5.29      | 7.24  | 9.72   | 7.39  |
|          | MAPE    | 2.73%    | 3.75% | 4.73% | 3.61% | 6.14%     | 8.80% | 13.40% | 9.06%  | 6.50%     | 9.33% | 14.08% | 9.55% |
| STGCN    | MAE     | 1.35     | 1.69  | 2.01  | 1.63  | 2.17      | 2.75  | 3.21   | 2.62   | 2.31      | 2.96  | 3.52   | 2.84  |
|          | RMSE    | 2.88     | 3.83  | 4.56  | 3.73  | 4.12      | 5.52  | 6.52   | 5.36   | 4.41      | 5.97  | 7.15   | 5.83  |
|          | MAPE    | 2.88%    | 3.85% | 4.74% | 3.69% | 5.19%     | 6.93% | 8.27%  | 6.58%  | 5.47%     | 7.43% | 9.05%  | 7.08% |
| GWNNet   | MAE     | 1.30     | 1.63  | 1.95  | 1.58  | 2.14      | 2.76  | 3.28   | 2.63   | 2.31      | 3.06  | 3.76   | 2.93  |
|          | RMSE    | 2.78     | 3.73  | 4.52  | 3.65  | 4.06      | 5.49  | 6.57   | 5.35   | 4.37      | 6.00  | 7.35   | 5.89  |
|          | MAPE    | 2.71%    | 3.66% | 4.63% | 3.52% | 5.12%     | 7.00% | 8.58%  | 6.65%  | 5.52%     | 7.23% | 10.03% | 7.53% |
| AGCRN    | MAE     | 1.37     | 1.70  | 1.99  | 1.63  | 2.86      | 3.41  | 3.73   | 3.22   | 2.38      | 3.11  | 3.64   | 2.94  |
|          | RMSE    | 2.93     | 3.89  | 4.64  | 3.78  | 5.45      | 6.68  | 7.31   | 6.35   | 4.54      | 6.29  | 7.41   | 6.07  |
|          | MAPE    | 2.95%    | 3.88% | 4.72% | 3.73% | 7.09%     | 8.60% | 9.56%  | 8.12%  | 5.80%     | 8.00% | 9.55%  | 7.53% |
| MTGNN    | MAE     | 1.34     | 1.67  | 1.97  | 1.60  | 2.15      | 2.74  | 3.21   | 2.62   | 2.28      | 2.96  | 3.59   | 2.84  |
|          | RMSE    | 2.84     | 3.79  | 4.55  | 3.70  | 4.10      | 5.53  | 6.58   | 5.37   | 4.37      | 6.01  | 7.34   | 5.89  |
|          | MAPE    | 2.80%    | 3.74% | 4.57% | 3.57% | 5.12%     | 6.81% | 8.19%  | 6.46%  | 5.46%     | 7.49% | 9.34%  | 7.17% |
| STemGNN  | MAE     | 1.44     | 1.93  | 2.57  | 1.92  | 2.41      | 3.19  | 4.27   | 3.16   | 2.55      | 3.40  | 4.51   | 3.36  |
|          | RMSE    | 3.12     | 4.38  | 5.88  | 4.46  | 4.38      | 6.02  | 7.99   | 6.10   | 4.68      | 6.46  | 8.51   | 6.54  |
|          | MAPE    | 3.08%    | 4.54% | 6.55% | 4.54% | 5.63%     | 8.07% | 8.19%  | 11.66% | 5.81%     | 8.38% | 12.11% | 8.42% |
| STNorm   | MAE     | 1.34     | 1.67  | 1.96  | 1.60  | 2.38      | 3.07  | 3.52   | 2.88   | 2.52      | 3.30  | 3.83   | 3.10  |
|          | RMSE    | 2.88     | 3.83  | 4.52  | 3.71  | 4.65      | 6.28  | 7.18   | 5.97   | 4.95      | 6.82  | 7.88   | 6.50  |
|          | MAPE    | 2.82%    | 3.75% | 4.62% | 3.60% | 5.73%     | 7.69% | 8.95%  | 7.17%  | 6.07%     | 8.37% | 9.93%  | 7.81% |
| ASTGNN   | MAE     | 1.35     | 1.69  | 1.98  | 1.62  | 2.18      | 2.91  | 3.72   | 2.82   | 2.20      | 2.92  | 3.72   | 2.83  |
|          | RMSE    | 2.86     | 3.82  | 4.57  | 3.73  | 4.22      | 5.88  | 7.56   | 5.87   | 4.22      | 5.97  | 7.55   | 5.88  |
|          | MAPE    | 2.83%    | 3.76% | 4.72% | 3.61% | 5.16%     | 7.37% | 9.96%  | 7.18%  | 5.07%     | 7.19% | 9.57%  | 6.98% |
| STID     | MAE     | 1.30     | 1.62  | 1.89  | 1.55  | 2.11      | 2.72  | 3.23   | 2.59   | 2.22      | 2.88  | 3.48   | 2.76  |
|          | RMSE    | 2.81     | 3.72  | 4.40  | 3.62  | 4.04      | 5.53  | 6.63   | 5.38   | 4.28      | 5.86  | 7.18   | 5.76  |
|          | MAPE    | 2.73%    | 3.68% | 4.47% | 3.51% | 5.17%     | 7.13% | 8.74%  | 6.76%  | 5.31%     | 7.42% | 9.31%  | 7.08% |
| MegaCRN  | MAE     | 1.28     | 1.60  | 1.88  | 1.58  | 2.13      | 2.75  | 3.28   | 2.60   | 2.22      | 2.94  | 3.71   | 2.83  |
|          | RMSE    | 2.72     | 3.68  | 4.42  | 3.61  | 4.16      | 5.49  | 6.51   | 5.35   | 4.21      | 6.00  | 7.36   | 5.85  |
|          | MAPE    | 2.67%    | 3.57% | 4.41% | 3.49% | 5.15%     | 6.80% | 8.25%  | 6.44%  | 5.13%     | 7.26% | 9.42%  | 7.21% |
| TD2MG2NN | MAE     | 1.25     | 1.55  | 1.85  | 1.50  | 2.07      | 2.64  | 3.13   | 2.52   | 2.20      | 2.82  | 3.36   | 2.70  |
|          | RMSE    | 2.64     | 3.50  | 4.22  | 3.44  | 3.94      | 5.31  | 6.38   | 5.08   | 4.24      | 5.75  | 6.90   | 5.61  |
|          | MAPE    | 2.62%    | 3.47% | 4.28% | 3.37% | 4.90%     | 6.57% | 8.06%  | 6.28%  | 5.25%     | 7.26% | 8.96%  | 6.91% |

- GWNNet [13]: GWNNet is the graph wavenet, constructing a dynamic adjacency matrix and employing one-dimensional dilated causal convolution to capture temporal and spatial dependencies, respectively.
- GMAN [41]: GMAN denotes the graph multi-attention network, which introduces attention mechanisms into spatial-temporal graph networks to capture crucial spatial-temporal features.
- MTGNN [38]: MTGNN represents the multivariate time graph neural network, optimizing the GWNNet structure for enhanced spatial and temporal dependency extraction capability.
- StemGNN [45]: StemGNN is the spectral spatial-temporal graph neural network, combining graph Fourier transform (GFT) for modeling inter-sequence correlations and discrete Fourier transform (DFT) for modeling temporal dependencies.
- ASTGNN [14]: ASTGNN denotes the attention-based spatial-temporal graph neural network, utilizing attention-based temporal feature extraction modules and conventional dynamic graph convolution for capturing spatial dependencies.
- STNorm [51]: STNorm introduces two normalization modules, namely, temporal normalization and spatial normalization, optimizing high-frequency components and local components in the original data, respectively.
- STID [52]: STID recognizes the non-distinguishability of samples in the spatial-temporal dimension as a critical bottleneck, proposing a simple and effective spatial-temporal prediction method based on a multilayer perceptron and spatial and temporal identity information.
- MegaCRN [49]: MegaCRN means the meta-graph convolutional recurrent network, which introduces a novel meta-graph learner and integrates it into RNN to capture spatial-temporal dependencies.

## 5.2. Experimental settings and metrics

In the experiments, we normalize the four datasets using the Z-Score method, following the velocity dataset partitioning approach established in prior research. The ratio of training, validation, and test sets for all velocity datasets is maintained at 7:1:2. Our objective is to predict future time series of length 12 on all datasets, and the results are presented in Tables 2 and 3. We compare the performance of baseline models and the Proposed Model at the 3rd, 6th, and 12th time slots, as well as the average of all 12 time slots, denoted in the @3, @6, @12, and avg columns in Tables 2 and 3. The best results are highlighted in bold, while the second-best results are underscored. The experiment is conducted on a server with the Ubuntu 22.04 operating system, NVIDIA GeForce RTX 4090 GPU, Intel Core i9-12900K CPU, Python version 3.8, and PyTorch version 1.10.0. During the training phase, we use the Adam optimizer with a learning rate of 0.001, a batch size of 64, and 200 epochs are employed, along with an early stopping strategy. MAE, MAPE, and RMSE are used as evaluation metrics during the validation and testing phases, defined by Eqs. (36), (37) and (38), and  $\hat{Y}_{i,j}$  is the ground truth. We use MAE as our loss function.

$$MAE = \frac{1}{N \times V} \sum_{i=1}^N \sum_{j=1}^V |Y_{i,j} - \hat{Y}_{i,j}| \quad (36)$$

$$MAPE = \frac{1}{N \times V} \sum_{i=1}^N \sum_{j=1}^V \left| \frac{Y_{i,j} - \hat{Y}_{i,j}}{Y_{i,j}} \right| \times 100\% \quad (37)$$

$$RMSE = \sqrt{\frac{1}{N \times V} \sum_{i=1}^N \sum_{j=1}^V (Y_{i,j} - \hat{Y}_{i,j})^2} \quad (38)$$

## 5.3. Performance study on public datasets

From Table 2, TD2MG2NN achieves nearly optimal prediction performance for all indicators in all situations on all public datasets,

demonstrating its superiority in long-term forecasting in generic conditions. In PEMS-BAY, TD2MG2NN shows on average improvements of 3.2%, 5.0%, and 4.0% in MAE, RMSE, and MAPE, respectively, compared to the best performing methods STID and MegaCRN. In PEMS7(M), TD2MG2NN outperforms STID and MegaCRN by an average of 2.7%, 5.6%, and 7.1% in MAE, RMSE, and MAPE, respectively. In PEMS7(L), TD2MG2NN is slightly inferior to MegaCRN in short-term prediction, but with average improvements of 2.2%, 2.6%, and 2.4% in MAE, RMSE, and MAPE respectively.

DCRNN and STGCN serve as fundamental spatial-temporal data forecasting models based on graph neural network (GNN). GWNet incorporates temporal convolution blocks with dilation coefficients and a graph convolutional network (GCN) with an adaptive adjacency matrix to capture temporal and spatial dependencies, respectively. Under the GWNet framework, MTGNN introduces multivariate time series to further enhance predictive performance. The introduction of attention mechanisms allows models to focus on global feature information, and ASTGNN achieves effective traffic prediction by incorporating attention mechanisms.

As the most recent traffic forecasting model, STNorm employs adaptive temporal and spatial feature matrices and demonstrates improved predictive performance through a simple multilayer perceptron model. STID, considering the inseparability of spatial-temporal dimensions, achieves optimal prediction results by constructing adaptive time and space feature matrices and utilizing a straightforward MLP model. MegaCRN introduces a meta-graph learning module and integrates it with RNN.

#### 5.4. Performance study on the rainy dataset

In contrast to normal conditions, conducting traffic speed prediction and traffic resilience assessment in heavy rainfall scenarios is more meaningful. The paper conducts a series of experiments on the PeMS-Rain dataset, which encompasses numerous heavy rainfall periods, to demonstrate the effectiveness of TD2MG2NN in predicting traffic speed during heavy rain. This serves as a foundation for resilience assessment. The comparative results of experiments with other baseline models are presented in Table 3. Overall, TD2MG2NN achieves the best prediction performance on PeMS-Rain. Compared to STID and MegaCRN, TD2MG2NN shows average improvements of 6.3%, 6.5% and 9.4% in MAE, RMSE and MAPE respectively, and 8.3%, 10.0%, and 8.3% in MAE, RMSE and MAPE. Among the GNN-based models, DCRNN, which is based on a basic RNN framework, shows inferior performance. STGCN, which uses one-dimensional convolution and conventional graph convolution to extract spatial-temporal dependencies, shows a significant improvement over DCRNN. Both GWNet and MTGNN introduce dynamic adjacency matrices and constructed dynamic graph structures to effectively capture dynamic spatial-temporal dependency features. MegaCRN, based on the proposed meta-graph learning mechanism, demonstrates the best predictive performance among GNN models. TD2MG2NN decomposes spatial-temporal speed data into seasonal and trend periodic data using multiscale decomposition. It uses two branch modules to finely process these two parts of the data, which effectively captures the abnormal fluctuations in speed data during heavy rain. In addition, the seasonal trend feature coding module introduces local-global temporal convolution, which further extracts the changing patterns of traffic speed during heavy rain. Finally, the model uses spatial decoding based on dynamic graph convolution to capture spatial dependencies in both spatial and temporal dimensions.

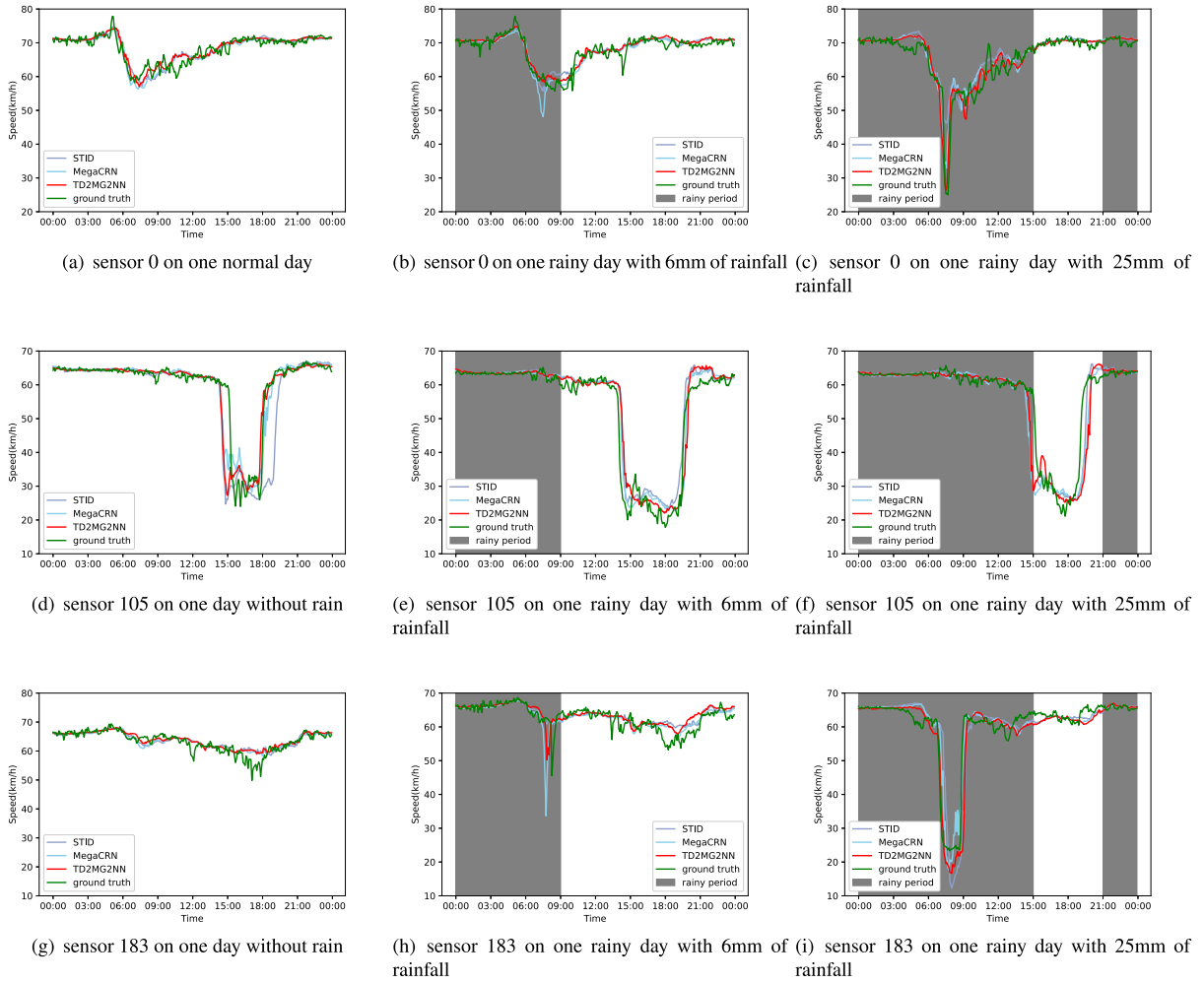
Additionally, to visually demonstrate the predictive advantage of TD2MG2NN over other models on the heavy rainfall datasets, we visualize the result of prediction on the PeMS-Rain. As Fig. 7 shows, the sensors selected in this paper are No. 0, No. 105 and No. 183, which are located in the center, south and northwest of the San Jose highway network respectively (marked with red boxes and red arrows). Sensor 0 is located at the intersection of four highways, and sensor 183

**Table 3**

Prediction results of TD2MG2NN and other state-of-the-art baselines on PeMS-Rain. The best results are highlighted in bold, while the second-best results are underscored.

| Model    | Metrics | PeMS-Rain    |              |              |              |
|----------|---------|--------------|--------------|--------------|--------------|
|          |         | @3           | @6           | @12          | avg          |
| DCRNN    | MAE     | 3.29         | 3.62         | 4.49         | 3.75         |
|          | RMSE    | 6.01         | 6.49         | 8.01         | 6.73         |
|          | MAPE    | 9.11%        | 9.80%        | 12.24%       | 10.17%       |
| STGCN    | MAE     | 1.57         | 2.05         | 2.53         | 1.96         |
|          | RMSE    | 3.07         | 4.15         | 5.02         | 4.02         |
|          | MAPE    | 3.40%        | 4.71%        | 6.03%        | 4.53%        |
| GWNet    | MAE     | 1.45         | 1.94         | 2.52         | 1.89         |
|          | RMSE    | <u>2.90</u>  | <u>3.98</u>  | 5.01         | <u>3.91</u>  |
|          | MAPE    | <u>3.05%</u> | <u>4.37%</u> | 5.98%        | <u>4.26%</u> |
| AGCRN    | MAE     | 1.53         | 2.03         | 2.52         | 1.94         |
|          | RMSE    | 3.08         | 4.23         | 5.13         | 4.02         |
|          | MAPE    | 3.35%        | 4.70%        | 6.03%        | 4.74%        |
| MTGNN    | MAE     | 1.57         | 2.05         | 2.60         | 2.00         |
|          | RMSE    | 3.13         | 4.29         | 5.28         | 4.47         |
|          | MAPE    | 3.60%        | 4.82%        | 6.37%        | 4.72%        |
| StemGNN  | MAE     | 1.55         | 2.11         | 2.85         | 2.08         |
|          | RMSE    | 3.15         | 4.46         | 5.91         | 4.51         |
|          | MAPE    | 3.38%        | 4.91%        | 6.97%        | 4.92%        |
| STNorm   | MAE     | 1.49         | 1.98         | 2.49         | 1.91         |
|          | RMSE    | 3.01         | 4.14         | 5.09         | 4.04         |
|          | MAPE    | 3.21%        | 4.53%        | 5.91%        | 4.40%        |
| ASTGNN   | MAE     | <u>1.43</u>  | <u>1.92</u>  | <u>2.42</u>  | <u>1.87</u>  |
|          | RMSE    | 2.98         | 4.09         | <u>4.85</u>  | 3.98         |
|          | MAPE    | 3.25%        | 4.61%        | 5.83%        | 4.39%        |
| STID     | MAE     | 1.48         | 1.97         | <u>2.42</u>  | 1.89         |
|          | RMSE    | 2.99         | 4.08         | 4.87         | 3.97         |
|          | MAPE    | 3.23%        | 4.58%        | 5.81%        | 4.38%        |
| MegaCRN  | MAE     | 1.46         | 1.97         | <u>2.42</u>  | 1.93         |
|          | RMSE    | 3.03         | 4.14         | 5.21         | 4.12         |
|          | MAPE    | 3.23%        | 4.64%        | <u>5.80%</u> | 4.33%        |
| TD2MG2NN | MAE     | <b>1.39</b>  | <b>1.81</b>  | <b>2.28</b>  | <b>1.75</b>  |
|          | RMSE    | <b>2.80</b>  | <b>3.79</b>  | <b>4.64</b>  | <b>3.69</b>  |
|          | MAPE    | <b>2.93%</b> | <b>4.09%</b> | <b>5.32%</b> | <b>3.94%</b> |

is located at the intersection of three highways, and sensor 105 is not at the intersection. We choose these three nodes to more comprehensively display the spatio-temporal characteristics of highway network speed. In Fig. 8, three different sensor nodes are selected to display the one-hour speed prediction curves of the proposed model, TD2MG2NN, and various state-of-the-art models in three different scenarios. The first column of sub-figures, labeled 8(a), 8(d), and 8(g), depicts the speed prediction curves under dry weather conditions. The second column, labeled 8(b), 8(e), and 8(h), presents speed prediction curves under a 6 mm rainfall scenario, while the third column, labeled 8(c), 8(f), and 8(i), displays speed prediction curves under 25 mm rainfall conditions. From the perspective of rainfall, it can be observed from the figures that the speed fluctuates more in the presence of 6 mm of rainfall compared to when there is no rain, with the most significant speed fluctuations occurring when there is 25 mm of rainfall. In these scenarios, each deep learning model demonstrates a greater ability to accurately capture the real speed change trend. The proposed TD2MG2NN model excels in capturing detailed trends, aligning with its outstanding predictive performance in terms of MAE, RMSE, and MAPE. Furthermore, as the amount of rainfall gradually increases, the overall predictive performance of the model deteriorates, although it remains at a high level. From a spatial perspective, Fig. 8 illustrates distinct patterns in the speed data of monitor nodes at various locations. On a rain-free day, sensor 0 experienced a significant drop in speed at 8 a.m., but maintained stability at other times. In contrast, sensor 105 exhibited a notable decrease around 5 p.m., with consistent speeds otherwise, while sensor 183 maintained a relatively stable speed profile throughout the day. When there was 6 mm of rainfall, node 0's speed decreased compared to non-rain conditions, sensor 105's speed remained relatively constant, and sensor 183 showed a declining trend. With 25 mm of rainfall, node 0 experienced a more substantial decrease



**Fig. 8.** One-hour prediction results of the sensors with ID 0, 105 and 183 on days without rain, rainy day with 6 mm of rainfall and rainy day with 25 mm of rainfall, using different SOTA models. Figs. 8(a), 8(d) and 8(g) illustrate the prediction effect of various SOTA models under normal weather; Figs. 8(b), 8(e) and 8(h) illustrate the prediction effect of various SOTA models under rainy weather with 6 mm of rainfall; Figs. 8(c), 8(f) and 8(i) illustrate the prediction effect of various SOTA models under rainy weather with 25 mm of rainfall. The green curves represent the actual speed data, the lavender curves depict the predictions of the STID model, the light blue curves show the predictions of the MegaCRN model, the red curves illustrate the predictions of the TD2MG2NN model proposed in this study, and the shaded area indicates the period of rainfall.

in speed than in non-rain conditions, sensor 105's speed remained stable, and sensor 183's speed declined sharply. Regardless of the location, TD2MG2NN's prediction performance consistently remains high, emphasizing its robustness.

### 5.5. Ablation experiments

To demonstrate the effectiveness of different modules in TD2MG2NN, a series of ablation experiments are conducted on two datasets, PeMSD7(L) and PeMS-Rain. Three variants of TD2MG2NN are considered:

- Remove multiscale Decomposition (w/o MD).
- Remove local-global Temporal Convolution (w/o local-global TemConv).
- Replace dynamic multi-granularity graph convolution with normal graph convolution (w/o DMG2Conv).

Ablation experiments are conducted in the same environment and training settings, and the results are presented in Table 4. The table illustrates the effectiveness of each module in TD2MG2NN and changes in the three metrics for each variant compared to the baseline are as follows: Compared to TD2MG2NN, For w/o MD, MAE, RMSE and MAPE decrease by approximately 6.3%, 2.1% and 4.5% on PeMSD7(L) and 4.6%, 1.3% and 4.0% on PeMS-Rain. The MD module effectively

separates the unstable seasonal and trend periodic features of time-series data during heavy rain by decomposing the data. This allows for the use of different time-series feature encoding modules. w/o L-G TemConv shows the reduction in MAE, RMSE and MAPE by roughly 5.9%, 3.2% and 4.6% on PeMSD7(L) and 5.2%, 2.9% and 4.5% on PeMS-Rain. This is due to the L-G TimeConv module can capture both short-term and long-term cyclic features, comprehensively representing the cyclic patterns of traffic. For w/o DMG2Conv, MAE, RMSE and MAPE are reduced by about 8.1%, 4.8% and 5.1% on PeMSD7(L) and 6.4%, 2.7% and 3.5% on PeMS-Rain. For spatial-temporal speed data, especially in heavy rainfall scenarios, the spatial dependency of speeds varies significantly at different time slices. The DMG2Conv module constructs different dynamic multi-level adjacency matrices at different time points, capturing the spatial dependency patterns of traffic speed evolving over time. In summary, all modules of TD2MG2NN contribute to enhancing the model's performance to some extent, aligning with the original design intentions. Additionally, for a more intuitive presentation of the ablation results for each module of TD2MG2NN, the experimental results are visualized as histograms, as shown in Fig. 9.

### 5.6. Resilience assessment results and analysis

Fig. 10 illustrates the resilience assessment results for the San Jose area, California, USA on two different heavy rainfall days, namely



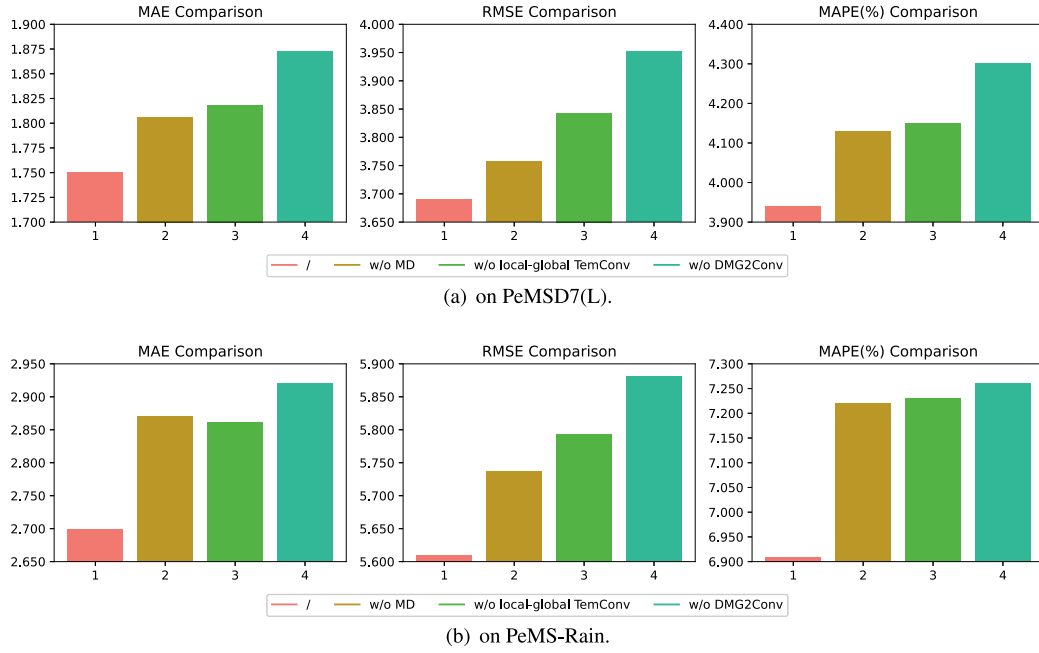


Fig. 9. Histogram of ablation study results for TD2MG2NN on PeMSD7(L) and PeMS-Rain.

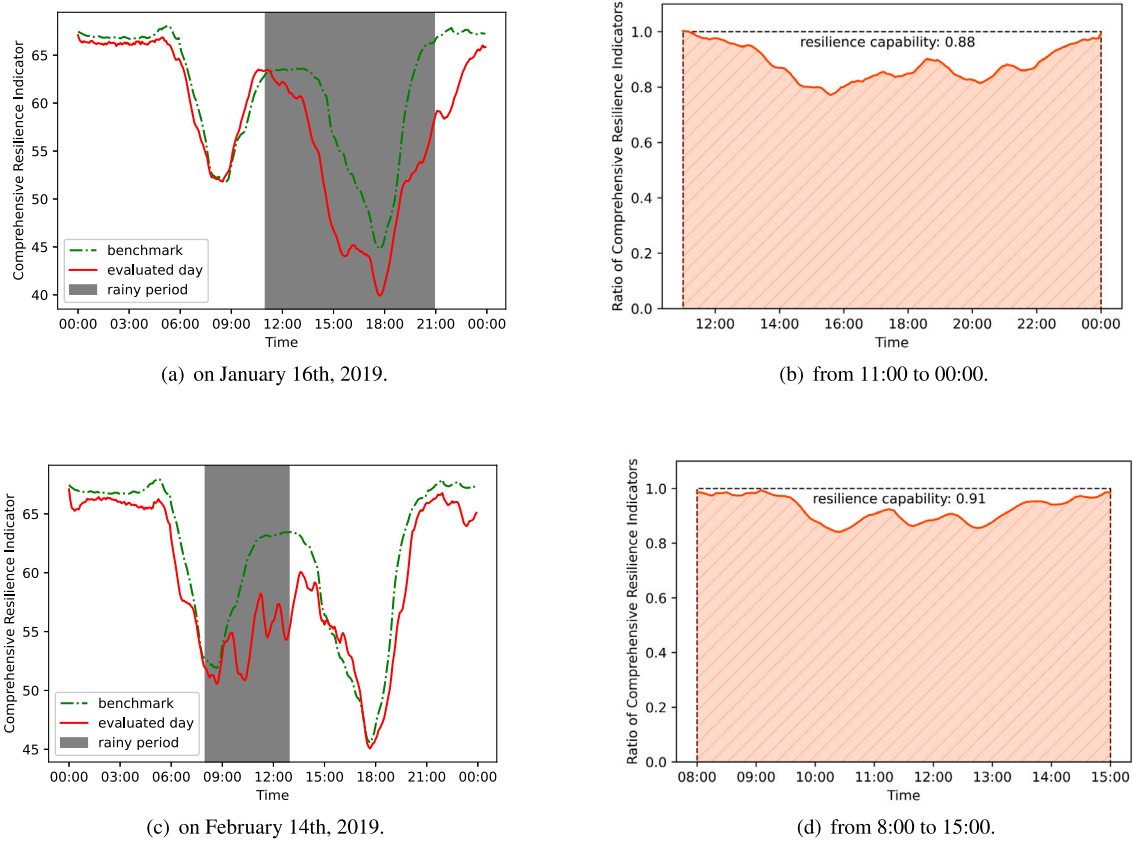
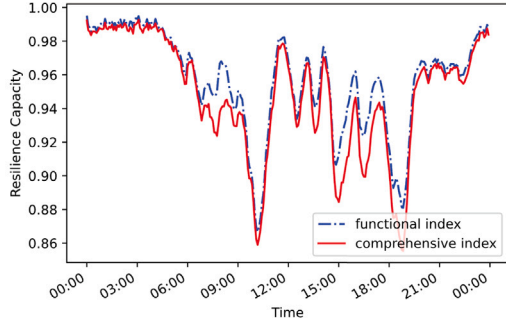


Fig. 10. Resilience assessment results during two different time periods, on January 16th, 2019 and February 14th, 2019. The green curves represent the corresponding benchmark of comprehensive resilience indicator calculated for non-rains days using Algorithm 1 in Section 4.1. The red curves in Figs. 10(a) and 10(c) indicate the variations in the comprehensive resilience indicator for the assessed rainy days of January 16, 2019, and February 14, 2019, which are obtained by Algorithm 2 in Section 4.1. The shaded areas indicate the periods of rainfall, which, according to 10(a) and 10(c), lasted from 12:00 to 21:00 and from 8:00 to 13:00, respectively. The orange curves in Figs. 10(b) and 10(d) respectively represent the ratio of the comprehensive resilience indicator to the corresponding benchmarks. On these two days of heavy rainfall, the resilience capability of the highway network in the San Jose area was 0.88 and 0.91.

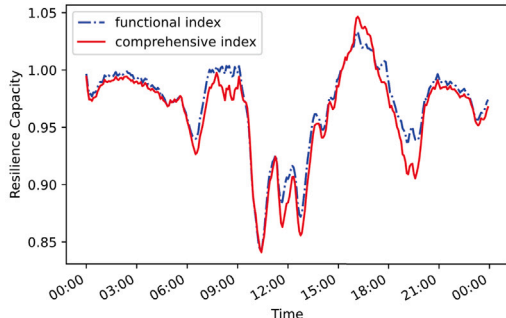
**Table 4**

Ablation results of TD2MG2NN on the PeMSD7(L) and PeMS-Rain datasets.

| w/o                      | PeMSD7(L) |      |       | PeMS-Rain |      |       |
|--------------------------|-----------|------|-------|-----------|------|-------|
|                          | MAE       | RMSE | MAPE  | MAE       | RMSE | MAPE  |
| w/o MD                   | 2.87      | 5.73 | 7.22% | 1.81      | 3.78 | 4.13% |
| w/o local-global TemConv | 2.86      | 5.79 | 7.23% | 1.82      | 3.84 | 4.15% |
| w/o DMG2Conv             | 2.92      | 5.88 | 7.26% | 1.87      | 3.95 | 4.30% |



(a) on February 4th in 2019.



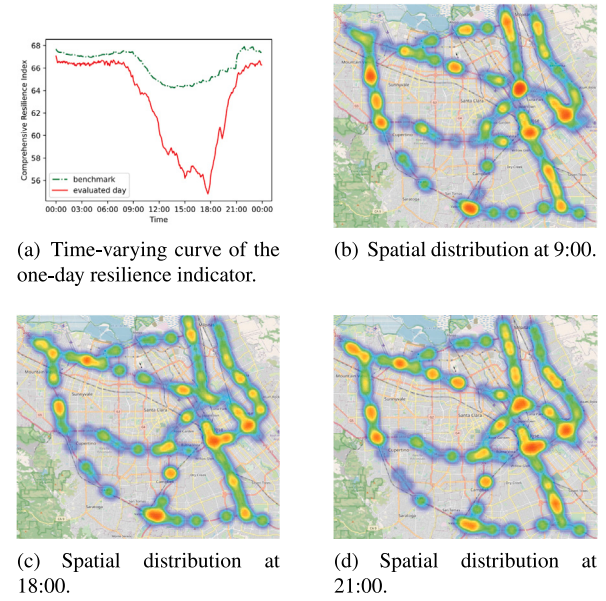
(b) on February 14th in 2019.

**Fig. 11.** Comparison results of resilience capability between functional indicator and comprehensive indicator on February 4th and 14th in 2019.

January 16, 2019, and February 14, 2019. Figs. 10(b) and 10(d) demonstrate that the end points of the resilience assessment occur three hours and two hours after the rain has ceased. This delay is attributed to the continuous impact of rainfall on travel conditions, requiring additional time for the road network's resilience to be restored. Resilience capability is quantified as the integral of the ratio of the resilience index from the starting point of assessment to the endpoint, relative to the integral when the ratio reaches 1. This calculation is visually represented by the ratio of the shaded area to the rectangle's area with a height of 1. A value of the resilience indicator ratio approaching 1 signifies a stronger state of the transportation system, while a value of the resilience capability near 1 indicates robust resilience within the transportation system. Detailed information on the calculation process can be found in Algorithm 3. During the heavy rainfall experienced on the aforementioned days, the resilience capability of the highway network in the San Jose area was measured at 0.88 and 0.91, as illustrated in Figs. 10(b) and 10(d). This value serves as a predictive tool for the traffic management department, enabling them to forecast the resilience of the road network and collect pertinent data to enhance future resilience recovery initiatives.

### 5.7. The advantage of proposed comprehensive resilience indicator

Fig. 11 shows the variation curves of the resilience capacity of the road network on two heavy rain days, namely February 4th, 2018, and February 14th, 2019, based on functional and comprehensive indicators. The figure suggests that the comprehensive resilience indicator

**Fig. 12.** Visualization of the comprehensive resilience assessment indicator's spatial-temporal distribution in the San Jose area on January 6th, 2019.

is more sensitive than the functional resilience indicator during the morning and evening peak hours, which provides a faster response to heavy rain events. The analysis shows that core nodes in the road network experience more significant speed reductions during heavy rainfall scenarios, while peripheral nodes remain relatively constant. The resilience indicator takes the structural characteristics of the road network into account, which increases the contribution weighting of core nodes to the overall resilience. This makes the resilience indicator of the whole road network more sensitive than the single functional resilience indicator.

### 5.8. Resilience assessment visualization

Fig. 12 illustrates the variation in resilience indicator for the San Jose area during the heavy rainfall day of January 6, 2019. Fig. 12(a) shows that rainfall in the San Jose area began to impact the roadway network's resilience at approximately 9:00 a.m. and continued until 9:00 p.m. when the impacts subsided. The three heat maps, labeled as 12(b), 12(c), and 12(d), display the spatial distribution of resilience in the region at three different points in time: 9:00 a.m. (when the rainfall impacts occur), 6:00 p.m. (when the rainfall has its greatest impact), and 9:00 p.m. (when the impact of rainfall gradually decreases), respectively. The resilience of the area near a point is indicated by the intensity of its color. The figures show that heavy rainfall leads to a decrease in the comprehensive resilience of network nodes. Core nodes experience a more rapid decline in resilience compared to other nodes. The visual heat map emphasizes the importance of core nodes in network resilience under heavy rainfall scenarios. This provides a theoretical basis for subsequent resilience enhancement efforts.

## 6. Conclusion

We develop a methodology for assessing the resilience of traffic road networks in real-time. This approach considers diverse states at different time points. It evaluates resilience by analyzing the evolution of the transportation system state, encompassing both historical states and future trends. In this methodology, an effective deep learning-based model (TD2MG2NN) for precise forecasting of speed is proposed. The model applies the architecture that separates seasonal and trend data using multiscale time series decomposition, providing more accurate

and explainable results for time series analysis. Besides, gated local-global temporal convolution is introduced for capturing short and long timing features comprehensively. Most importantly in TD2MG2NN, we propose the dynamic multi-granularity graphs for extracting multi-level and time-varying spatial dependencies, providing a more comprehensive perspective for understanding spatial relationships and temporal evolution. Besides, we construct a comprehensive spatial-temporal resilience indicator, which is formulated by integrating both functional and structural features of the urban road network. This indicator successfully reflects spatial-temporal changes in the network before, during, and after disruptions caused by heavy rainfall. In future work, we will further explore the impact of various regulatory measures and driving factors on the enhancement of traffic system resilience.

### CRedit authorship contribution statement

**Di Zang:** Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition. **Yongjie Ding:** Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation. **Jiayi Zhao:** Software, Resources, Data curation. **Keshuang Tang:** Writing – review & editing, Supervision, Conceptualization. **Hong Zhu:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Data availability

Data will be made available on request.

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